

# $A \cup \phi = A$ $A \cap U = A$	Identity law	# $A \cap A = A$ $A \cup A = A$	Idempotent law
# $A \cup U = U$ $A \cap \phi = \phi$	Domination law	# $\overline{(\overline{A})} = A$	Complementation law
# $A \cup \overline{A} = U$ $A \cap \overline{A} = \phi$	Complement law	# $\overline{(A \cap B)} = \overline{A} \cup \overline{B}$ $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$	De Morgan law
		# $\overline{A} \cap (A \cup B) = \overline{A}$ $A \cup (A \cap B) = A$	Absorption law

Ex Prove that $A - B = A \cap \overline{B}$

Pf Let $x \in (A - B) \Leftrightarrow (x \in A) \wedge x \notin B$ ()
 $\Leftrightarrow (x \in A) \wedge (x \in \overline{B})$ ()
 $\Leftrightarrow x \in (A \cap \overline{B})$ (Defⁿ intersection)

$A - B = A \cap \overline{B}$

Ex Let A, B, C be sets. $\overline{A \cup (B \cap C)} = (\overline{A} \cap \overline{B \cap C})$

Sol $\overline{A \cup (B \cap C)} = \overline{A} \cap \overline{(B \cap C)}$ (De Morgan law)
 ")

(h) $A \subseteq (A \cup B)$

Ex Let $x \in A$
 $\Rightarrow (x \in A) \vee (x \in B)$ (Addition)

Sol ($\Rightarrow x \in (A \cup B)$

$A \subseteq (A \cup B)$

Ex 16 a) $(A \cap B) \subseteq A$ (b) $A \subseteq (A \cup B)$ (c) $(A - B) \subseteq A$

d) $A \cap (B - A) = \phi$ e) $A \cup (B - A) = A \cup B$

(a) $(A \cap B) \subseteq A$

Let $x \in (A \cap B)$

$\Rightarrow (x \in A) \wedge (x \in B)$

$\Rightarrow x \in A$ (Simplification)

$(A \cap B) \subseteq A$

2.3 Functions

Set $\Rightarrow$ Cartesian Product $\Rightarrow$ Relation $\Rightarrow$ Function

Cartesian Product (Cross Product) | # Relation

Let A & B be sets

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

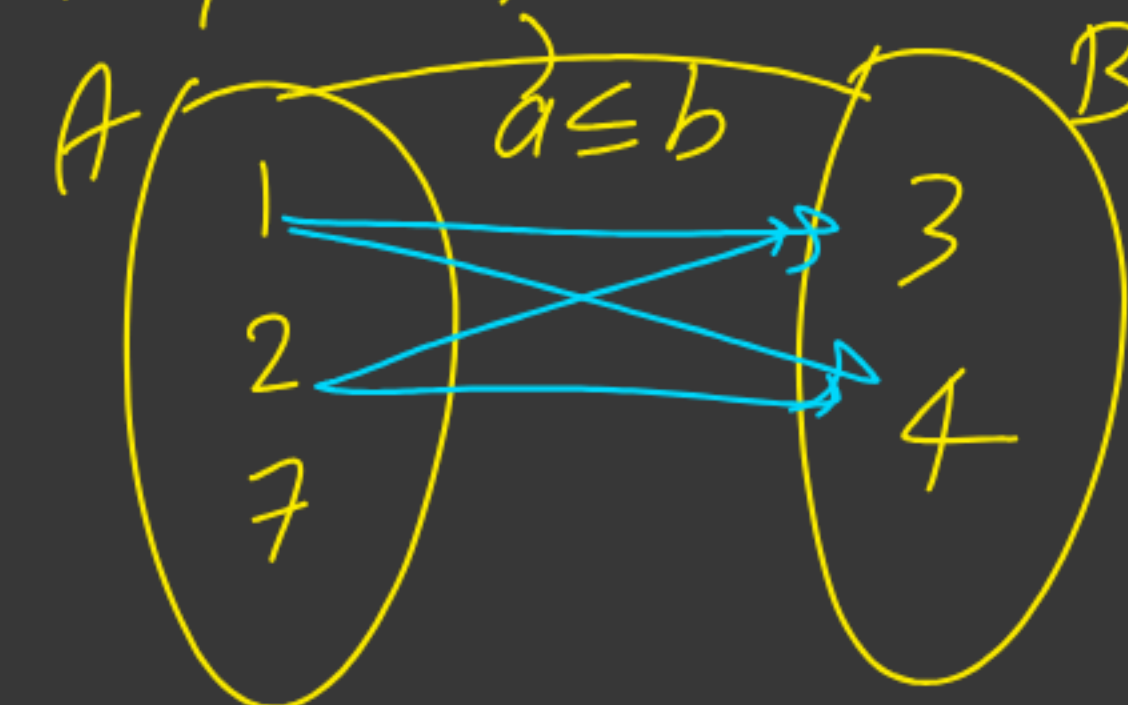
Ex: $A = \{1, 2\}$ $B = \{3, 4\}$

$$A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$$

Ex: $A = \{1, 2, 7\}$ $B = \{3, 4\}$

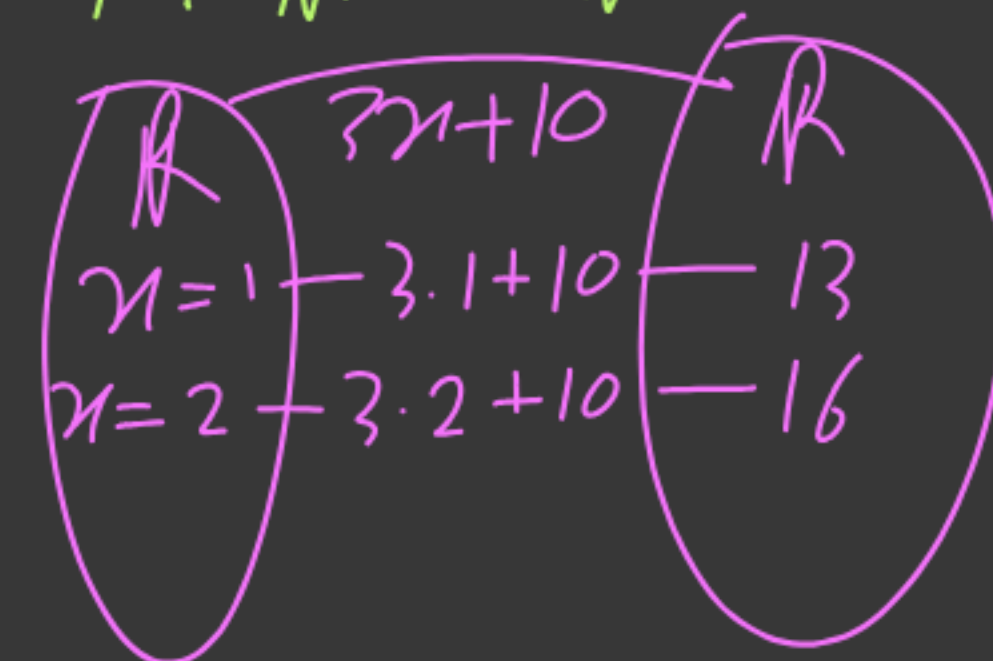
$$R = \{(a, b) \mid a \leq b, a \in A, b \in B\}$$

$$= \{(1, 3), (1, 4), (2, 3), (2, 4)\}$$



Ex: $f(x) = 3x + 10$

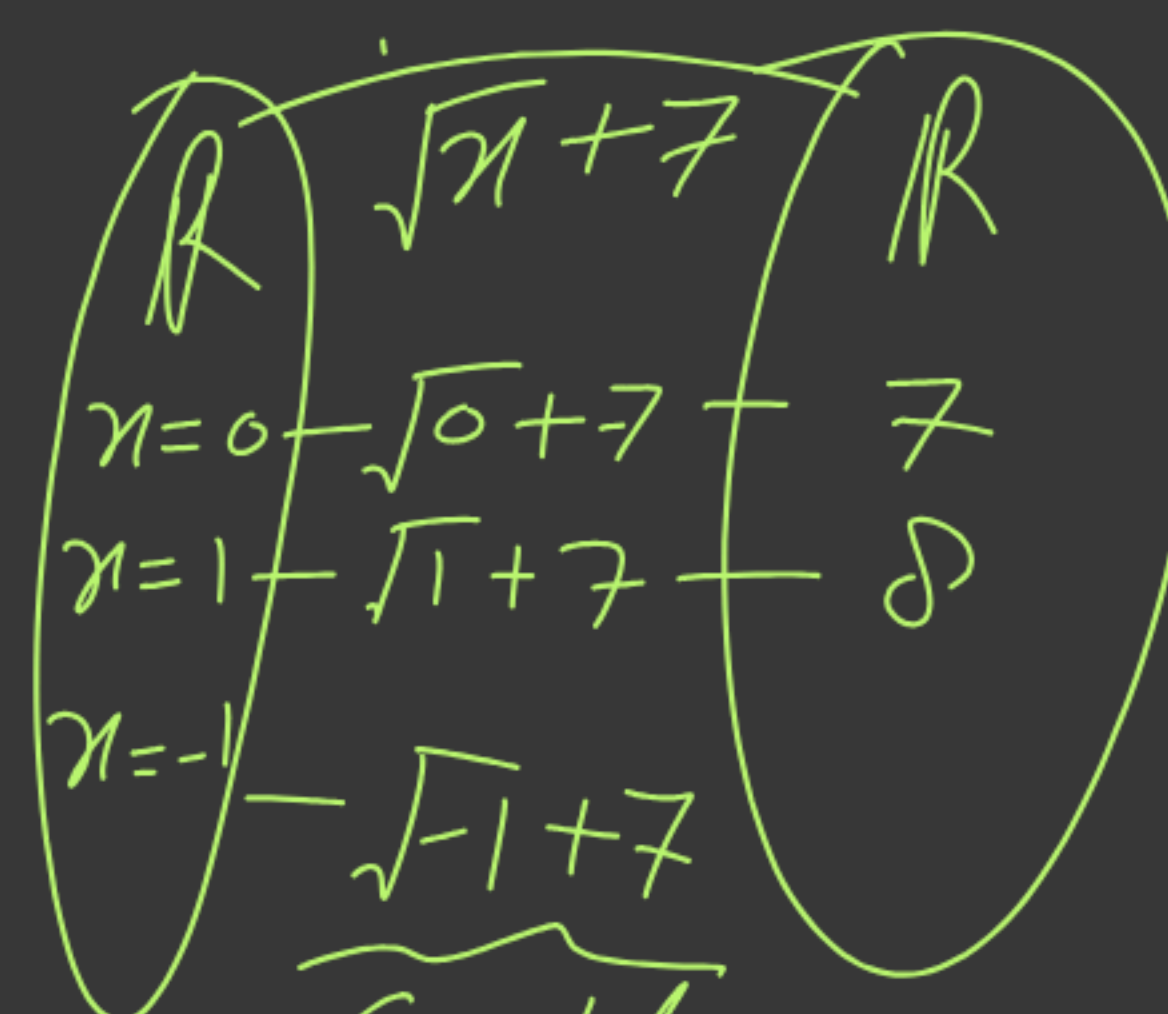
$$f: \mathbb{R} \rightarrow \mathbb{R}$$



① ✓
② $x_1 \rightarrow 3x_1 + 10 \rightarrow y_1$
(X) $\rightarrow y_2$

Ex: $f(x) = \sqrt{x} + 7$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$



Complex No.

① X

f is not a f

Domain = $\mathbb{R} - (-\infty, 0)$ $\mathbb{R}^-$ | $\mathbb{R}^+$

$$\underline{\underline{f(c) \geq f(x), \forall x \in D.}}$$

Ch:-4. Applications of Differentiation.

4.1 Maximum and Minimum Values.

Max or Min	<u>$f(x)$</u>	Absolute	Max / Min $\rightarrow$ Global
		Local	Max / Min.
Domain $- D$			

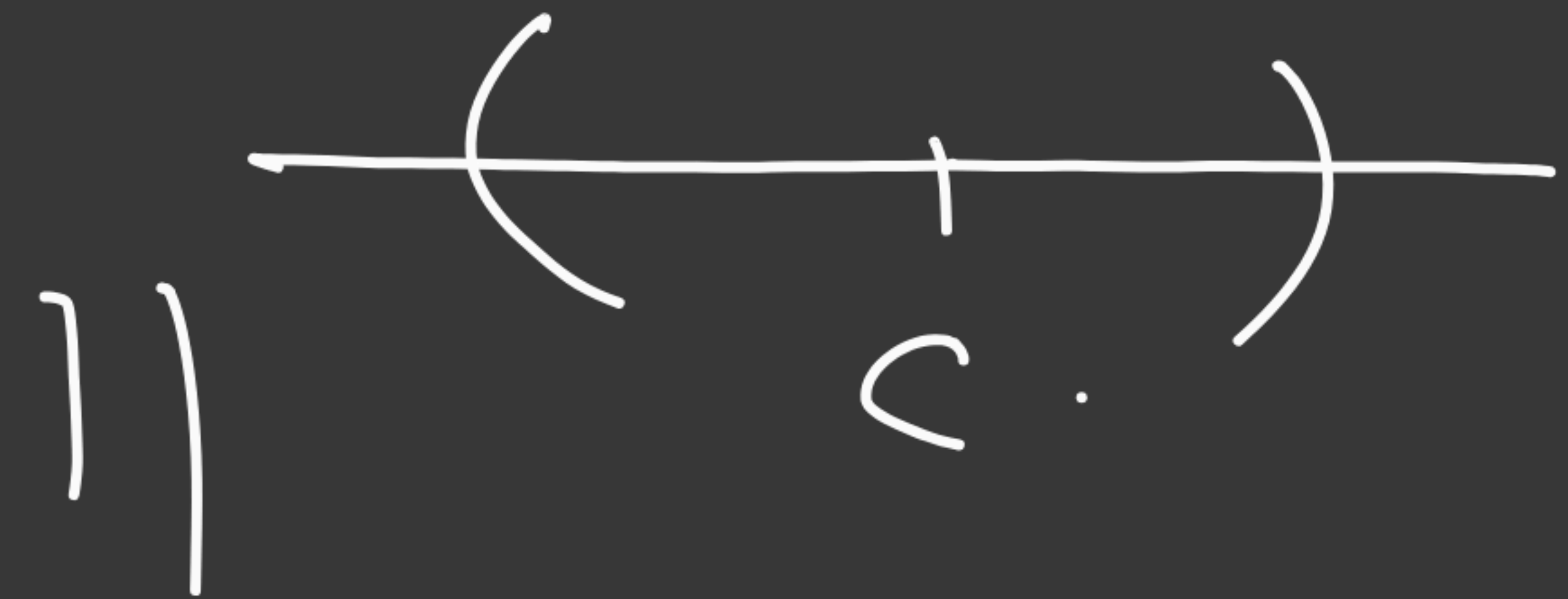
Let c be a no. belonging to the domain D of the funⁿ f and $f(c)$ is said to be the abs. max value of f if

min. $f(c) \geq f(x), \forall x \in D.$

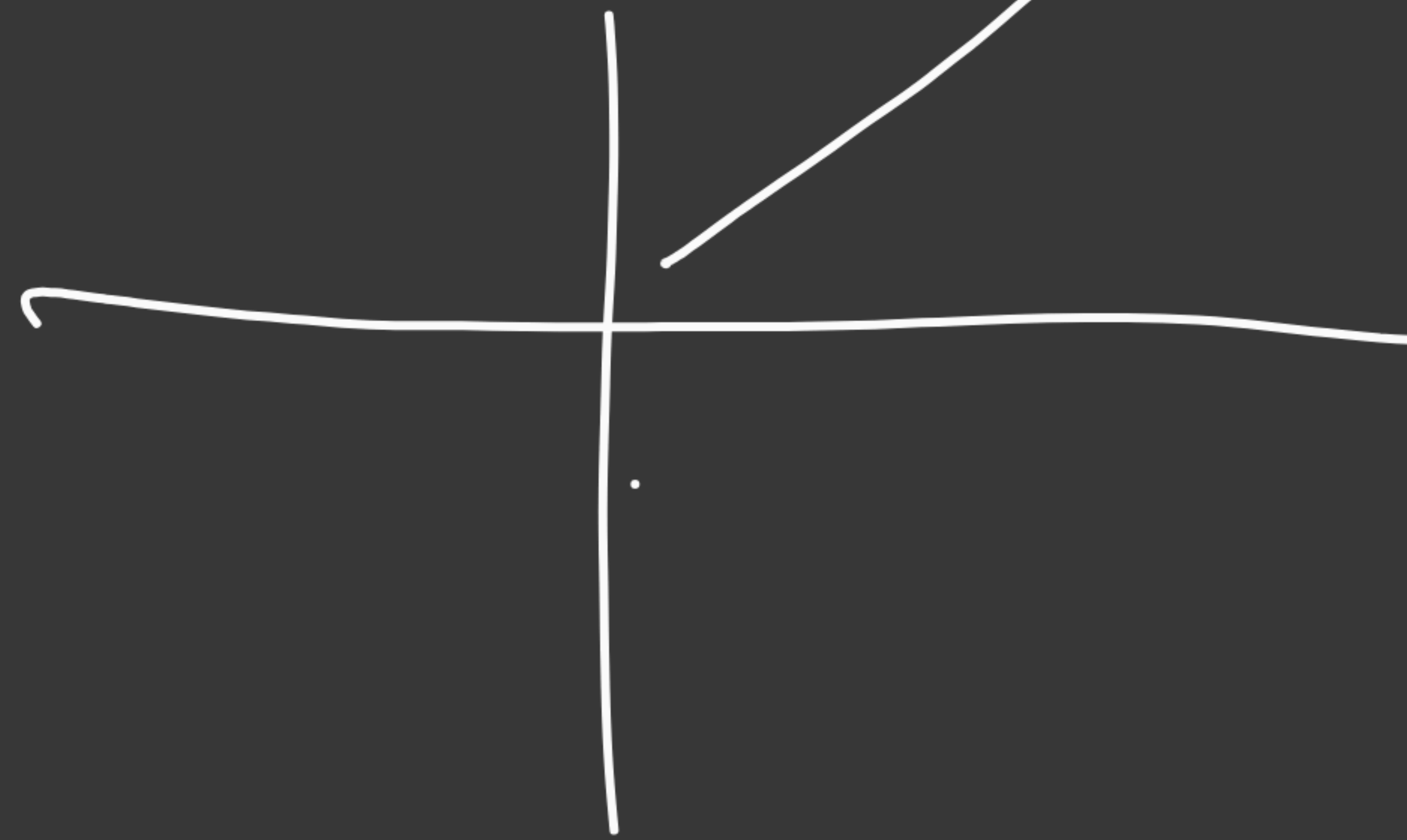
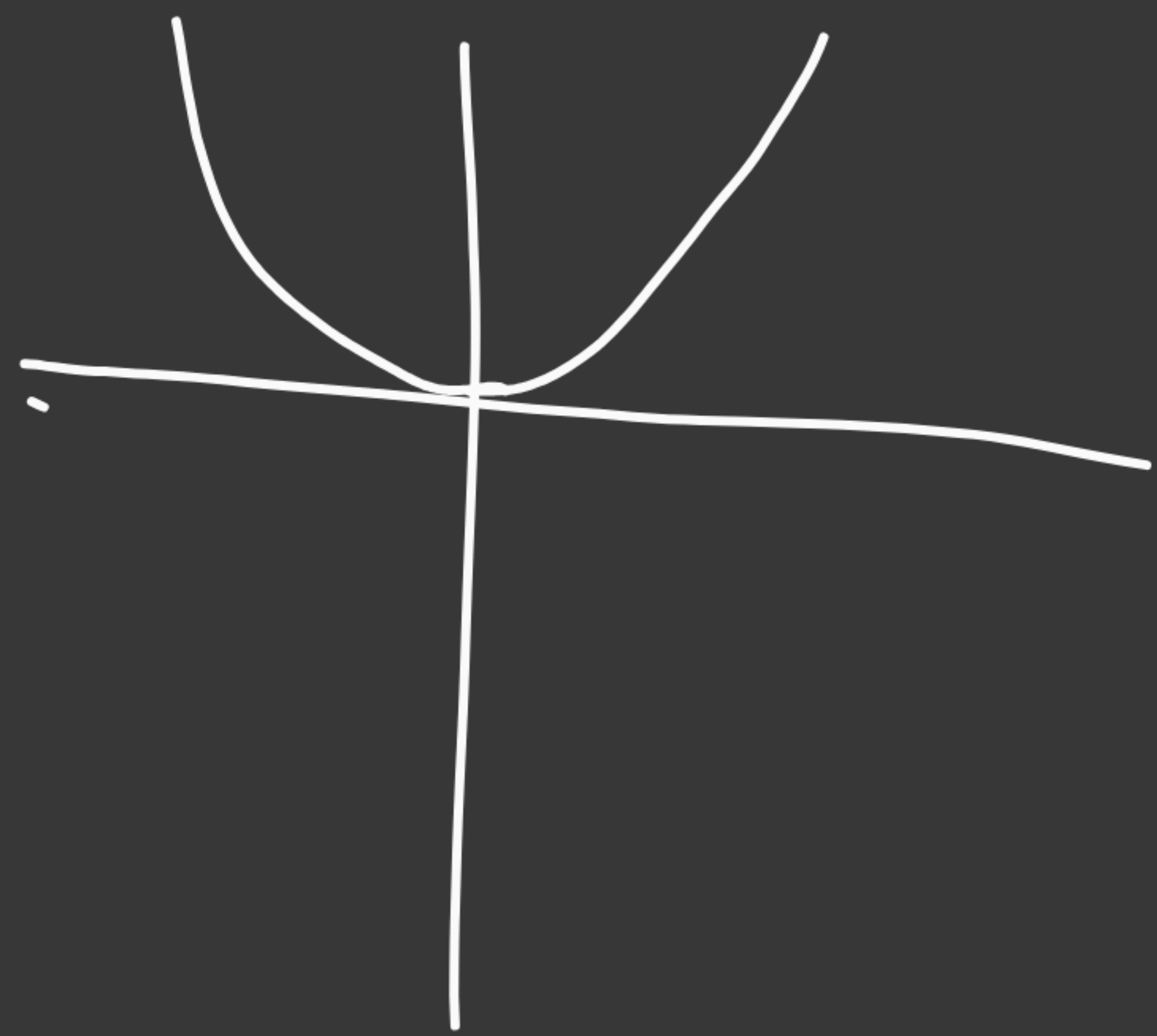
$\rightarrow f(c) \leq f(x), \forall x \in D.$

$f(c)$ is a local $\overset{\text{min}}{\text{max}}$ ^{value} of the funⁿ $f(x)$
if $\underline{f(c) \geq f(x)}$, for all x near c .

$$f(c) \leq f(x)$$



The Extreme Value Th:- If f is continuous on a closed interval $[a, b]$ then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some nos. c and d in $[a, b]$.

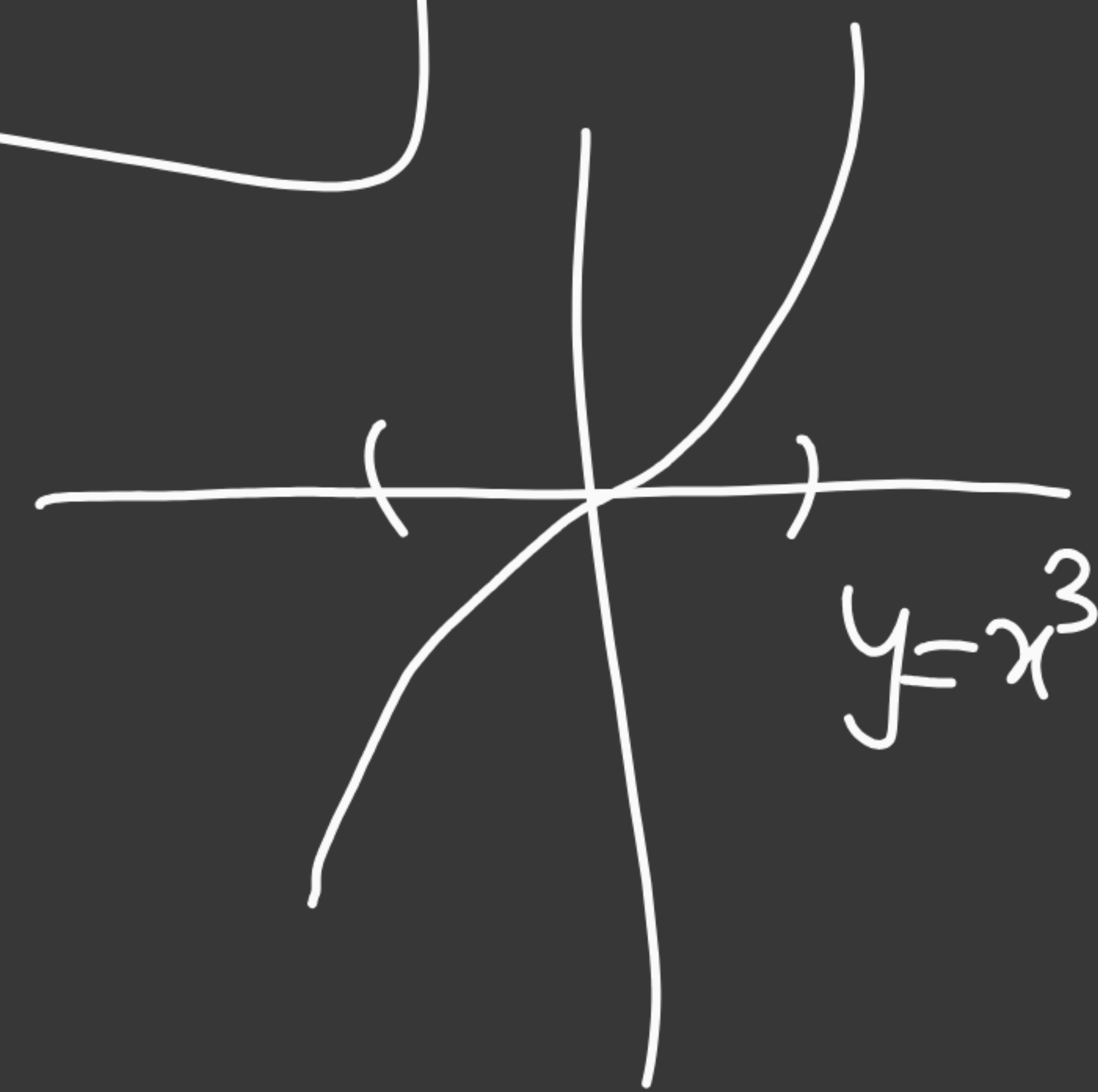


1. Fermat's Th:- If f has a local maximum/minimum at $\underline{c}$, and if $f'(c)$ exists then $f'(c) = 0$.

local max or min $\Rightarrow$ $f'(c) = 0$

Ex: provided $f'(c)$ exists

$$f(x) = x^3 \Rightarrow f'(x) = 3x^2 = 0 \Rightarrow \boxed{x = 0}$$



$$x^2 - x - 6 = 0 \Rightarrow x = -2, x = 3.$$

Defⁿ:- A critical number of a fuⁿ f is a no. c in the domain of the fuⁿ $f(x)$ such that either $f'(c) = 0$ or $f'(c)$ does not exist.

31) Find the critical nos. of the fuⁿ $f(x) = 2x^3 - 3x^2 - 36x$.

Solⁿ:- $f'(x) = 6x^2 - 6x - 36$

Equating $f'(x) = 0$, $6x^2 - 6x - 36 = 0$.
 $\Rightarrow 6(x^2 - x - 6) = 0$.

$h(t)$ is not differentiable at $t=0 \Rightarrow t=0$ is a critical no.

$$37) \quad h(t) = t^{3/4} - 2t^{1/4}.$$

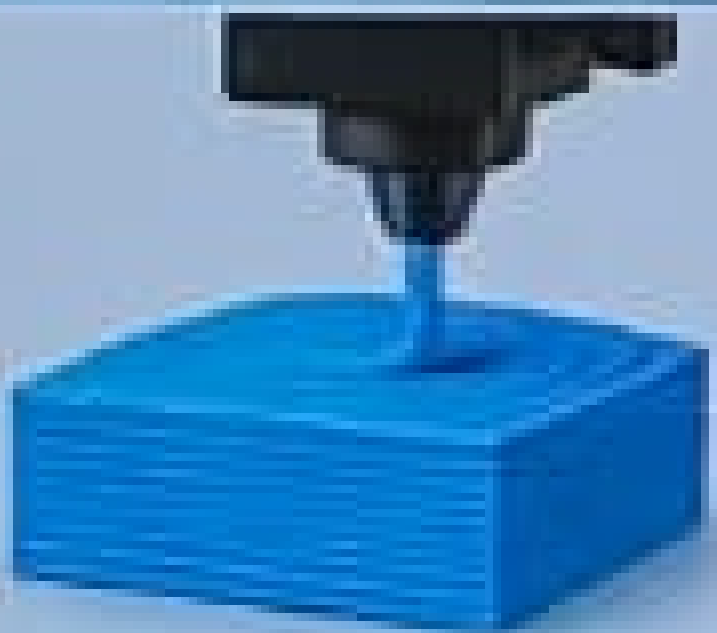
$$h'(t) = \frac{3}{4} \frac{1}{t^{1/4}} - 2 \times \frac{1}{4} \frac{1}{t^{3/4}} = \frac{1}{4} \left[\frac{3}{t^{1/4}} - \frac{2}{t^{3/4}} \right]$$

$$= \frac{1}{4} \left(\frac{3t^{2/4} - 2}{t^{3/4}} \right) = \frac{1}{4} \left(\frac{3\sqrt{t} - 2}{t^{3/4}} \right)$$

$$\text{Equating } h'(t) = 0 \Rightarrow \frac{1}{4} \left(\frac{3\sqrt{t} - 2}{t^{3/4}} \right) = 0 \Rightarrow 3\sqrt{t} - 2 = 0 \\ \Rightarrow \sqrt{t} = \frac{2}{3} \Rightarrow t = \frac{4}{9}.$$

Defⁿ :— If f has a local maximum
or minimum at c , then c is a critical
no. of f .

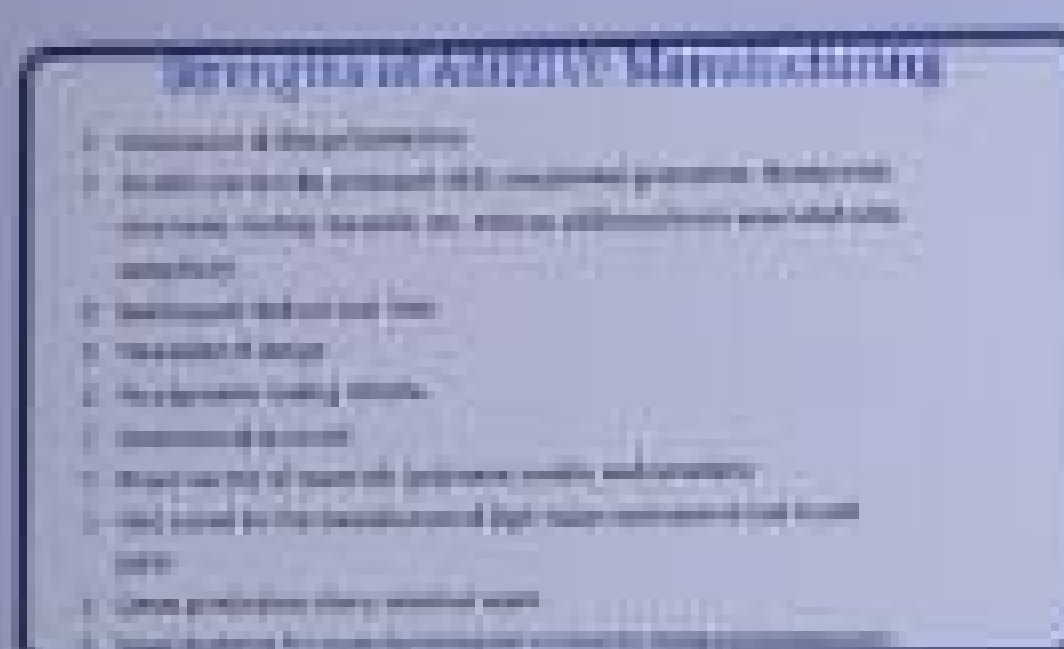
Summary of AM Technologies

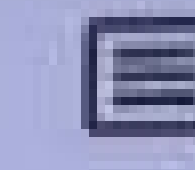
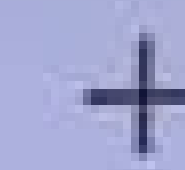
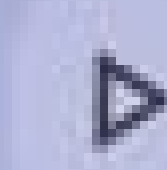
Technique	Brief Description	Typical Material Usage	Common Applications
1 Fused Deposition Modeling (FDM) 	Builds parts by heating and extruding a thermoplastic filament layer by layer.	- PLA (Polylactic Acid) - ABS (Acrylonitrile Butadiene Styrene) - PETG (Polyethylene Terephthalate Glycol) - TPU (Thermoplastic Polyurethane)	- Prototypes and functional parts - Low-cost production - Consumer products, tooling, fixtures
2 Stereolithography (SLA) 	Uses a UV laser to cure liquid photopolymer resin, layer by layer.	- Photopolymer resins - Epoxy-based resins	- High-precision prototypes - Jewelry, dental models - Medical and design models
3 Selective Laser Sintering (SLS) 	Uses a laser to fuse powdered material, layer by layer.	- Nylon (Polyamide) - TPU - Metal powders (e.g., Titanium, Steel, Aluminum)	- Functional end-use parts - Aerospace, automotive - Industrial tooling
4 Direct Metal Laser Sintering (DMLS) 	Uses a laser to fully melt and fuse metal powder, layer by layer.	- Titanium alloys - Aluminum alloys - Stainless steel - Nickel alloys	- Aerospace components - Medical implants - High-strength, heat-resistant parts
5 Binder Jetting 	Uses a liquid binder to join powder particles, layer by layer.	- Sand (foundry) - Metal powders (with post-processing) - Ceramic powders	- Tooling and molds - Investment casting (foundry) - Complex metal parts
6 Material Jetting 	Uses inkjet print heads to deposit photopolymer or wax droplets, which are cured layer by layer.	- Photopolymer resins - Wax (for investment casting) - Composites (multi-material)	- Full-color prototypes - Investment casting patterns - Medical and dental models
7 Laminated Object Manufacturing (LOM) 	Builds parts by laminating sheets of material and cutting the shape with a laser or blade.	- Paper - Plastic films (e.g., ABS, acrylic) - Composite sheets	- Concept models - Architectural models - Large parts and tooling



Strengths of Additive Manufacturing

- Elimination of design limitations
- Enables parts to be generated with complicated geometries: Honeycomb structures, cooling channels, etc., with no additional costs associated with complexity
- Build speed: Reduce lead time
- Flexibility of design
- No expensive tooling criteria
- Dimensional accuracy
- Broad variety of materials (polymers, metals, and ceramics)
- Well suited for the manufacture of high-value replacement and repair parts
- Green production, clean, minimal waste
- Small footprint for manufacturing and constantly rising equipment costs





Applications and Limitations of AM

Applications

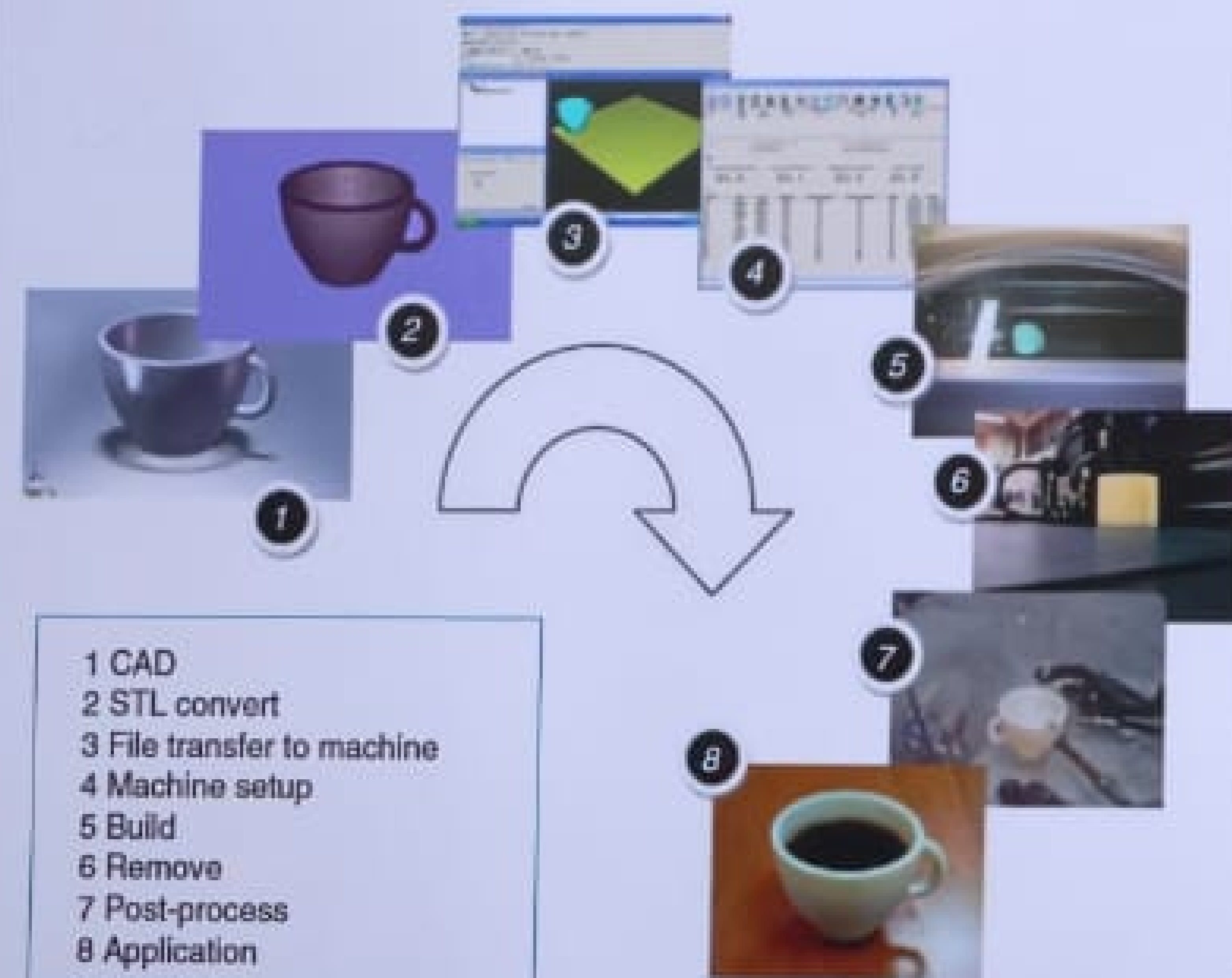
- ✓ Lower energy consumption
- ✓ Less waste
- ✓ Less dedicated tooling
- ✓ Reduced development costs and time to market
- ✓ Innovative designs and geometries
- ✓ Part consolidation (fewer parts with more complex designs)
- ✓ Customization of parts (e.g., for medical implants, specialty repair parts, parts where other manufacturing facilities are not available, such as on ships or in space)

Limitations

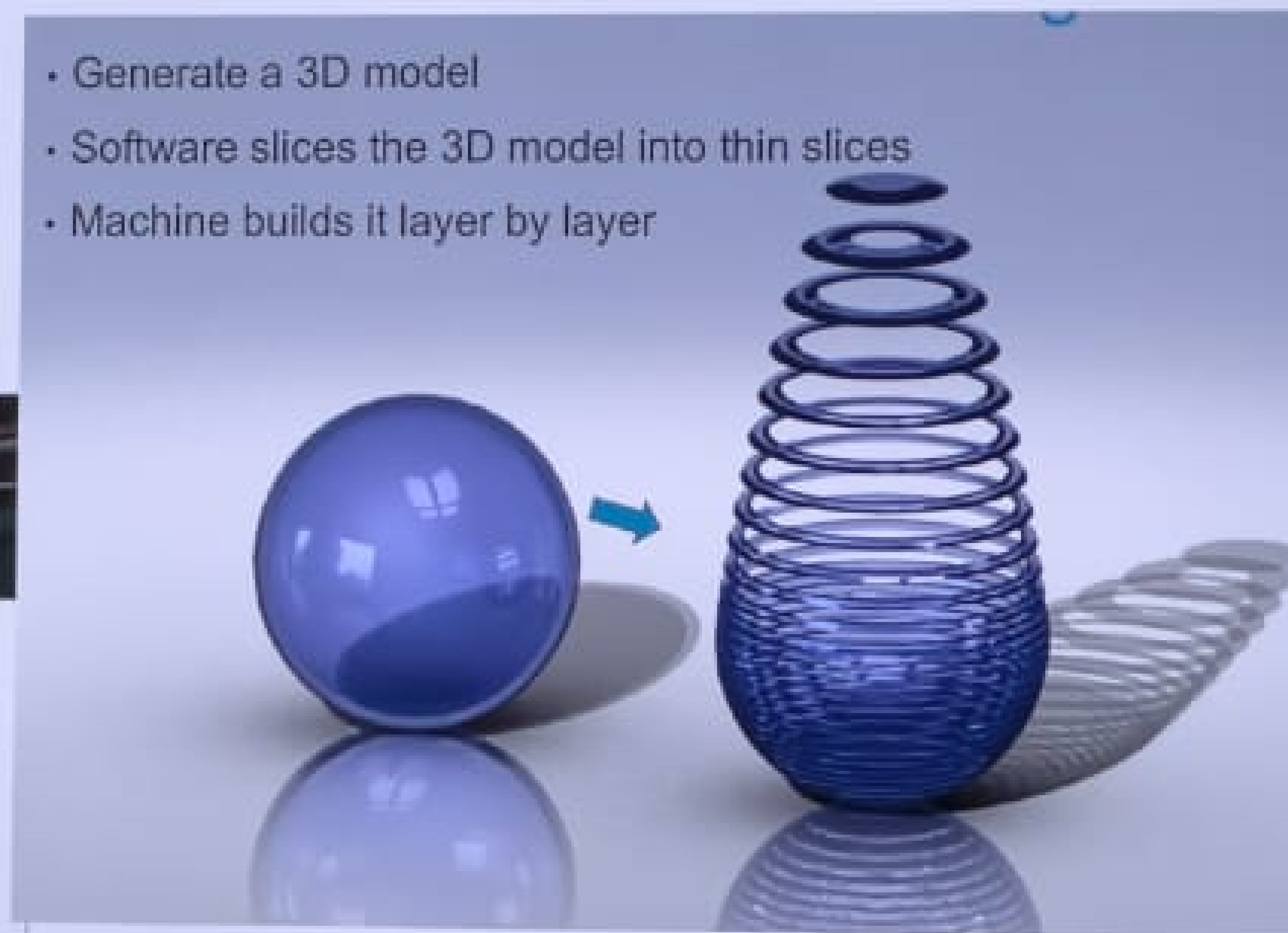
- Surface roughness
- Low density, porosity
- Lack of data on the end-use properties of parts to be produced (e.g., thermal and chemical stability, strength, etc.)
- Limited to relatively small parts



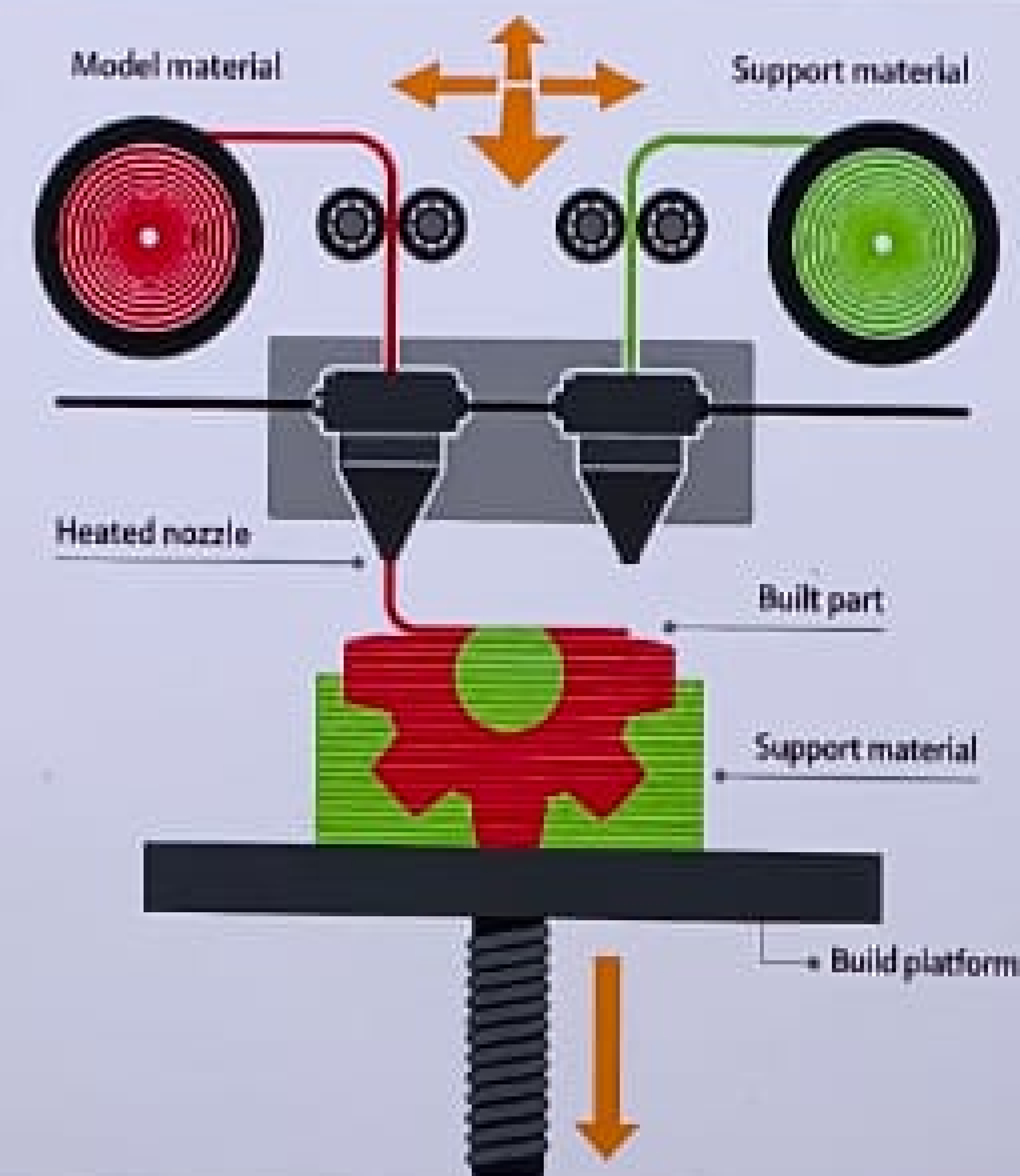
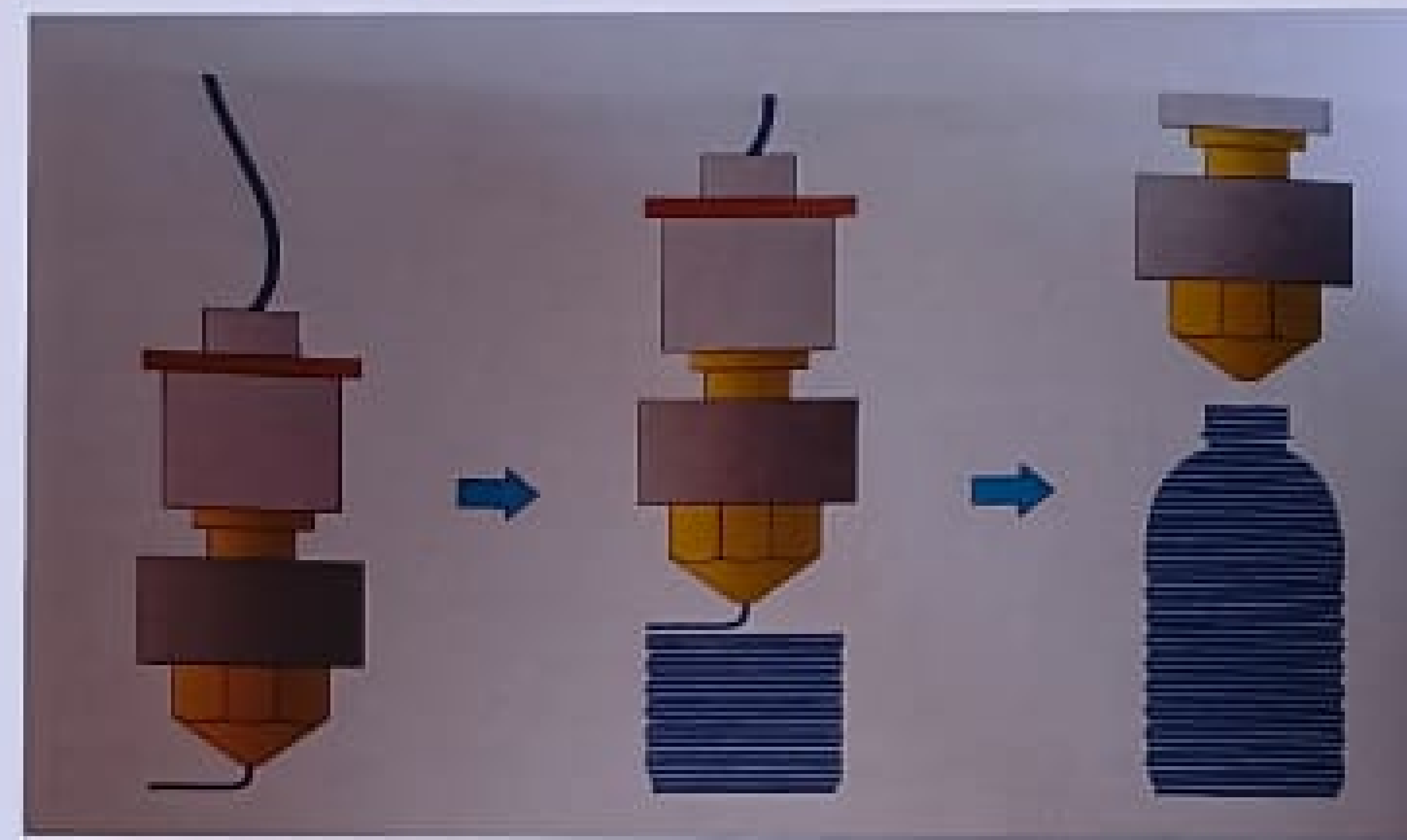
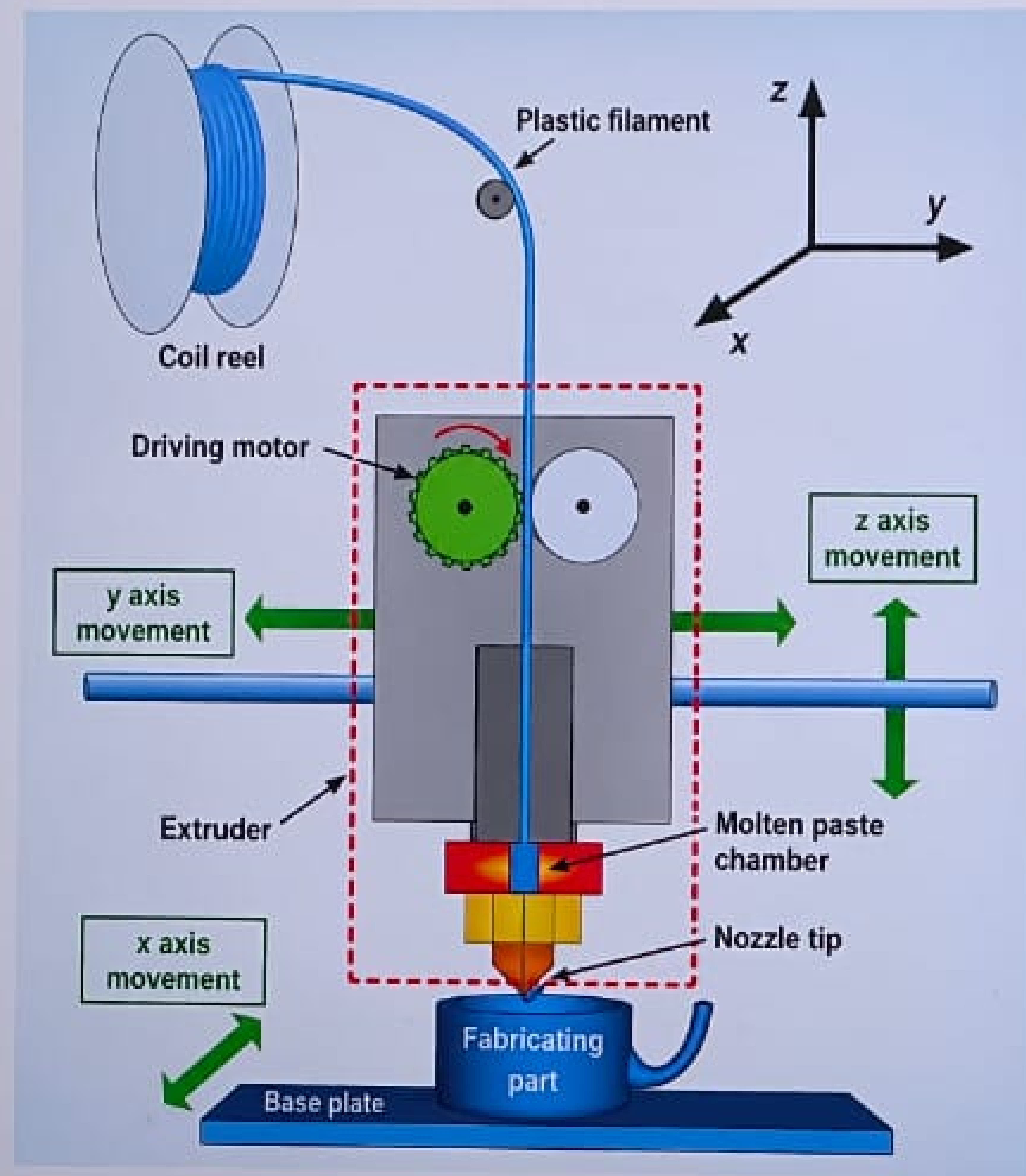
AM Process Chain



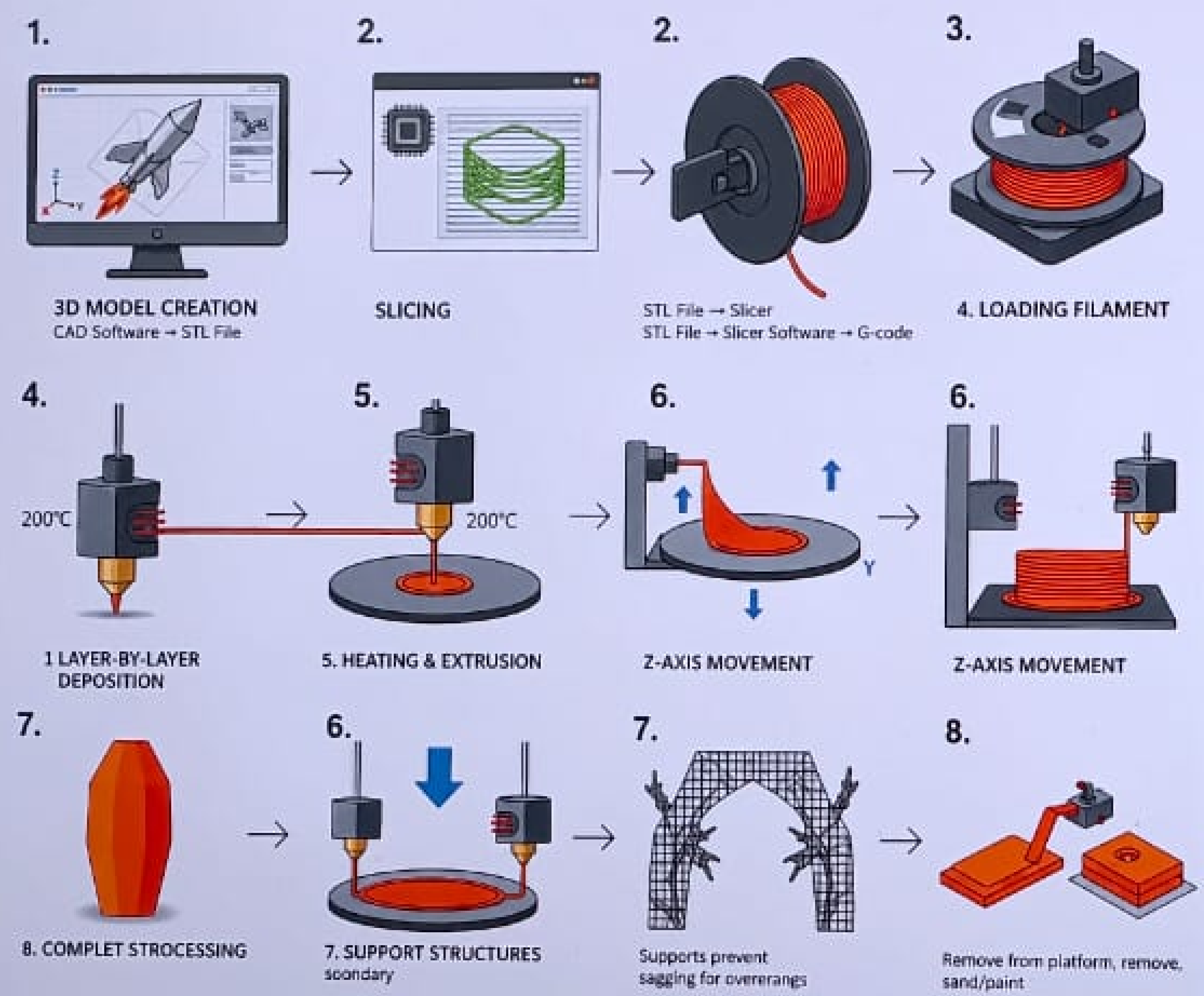
- Generate a 3D model
- Software slices the 3D model into thin slices
- Machine builds it layer by layer



Fused Deposition Modeling (FDM)



FDM Process Methodology



As deposited FDM



Removal of support material

FDM Process steps

Step 1: CAD Model Creation

Step 2: Conversion into an STL
File

Step 3: Slicing Process

Step 4: Loading Filament

Step 5: Heating and Extrusion

Step 6: Layer – by – layer
Deposition

Step 7: Cooling and
Solidification

FDM Process

1. CAD Model Creation...

The process begins with creating a 3D design using CAD software such as:

- ✓ AutoCAD
- ✓ SolidWorks
- ✓ CATIA
- ✓ Fusion 360
- ✓ Tinkercad

The model defines the shape and dimensions of the object.

2. Conversion into STL File

- The CAD model is converted into an **STL** (Standard Tessellation Language) file.
- ✓ The STL format represents the surface geometry of the object using triangles.
- ✓ It is the standard file format used in 3D printing.

FDM Process

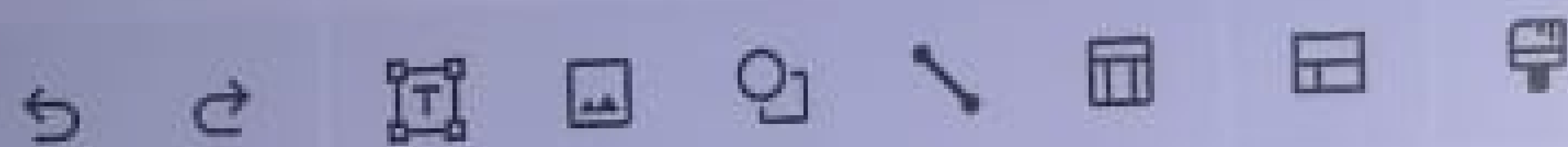
3. Slicing Process...

The STL file is imported into slicing software such as:

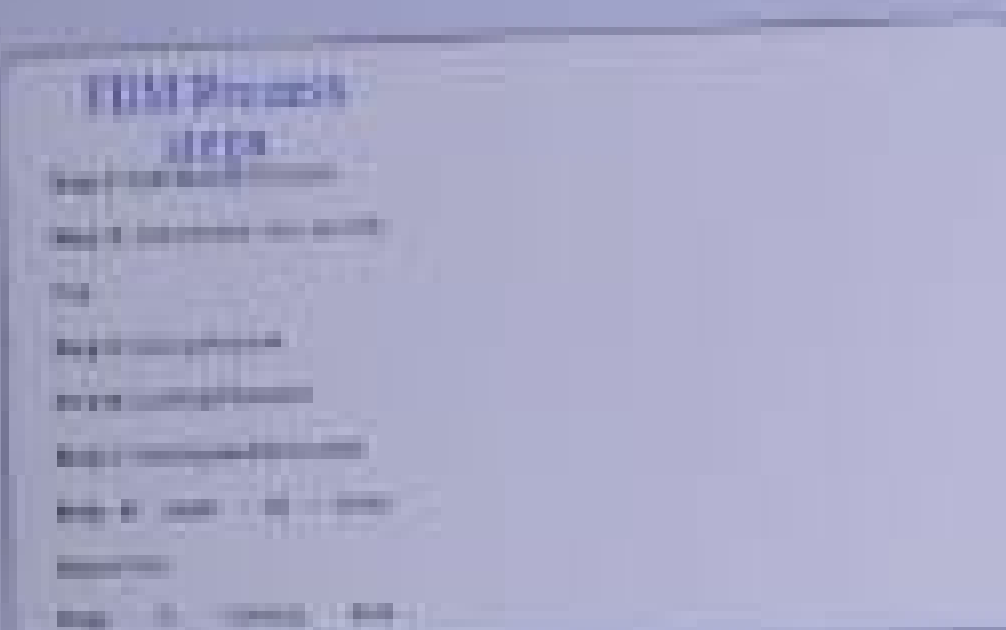
- ✓ Cura
- ✓ Prusa Slicer
- ✓ Simplify3D

The slicer

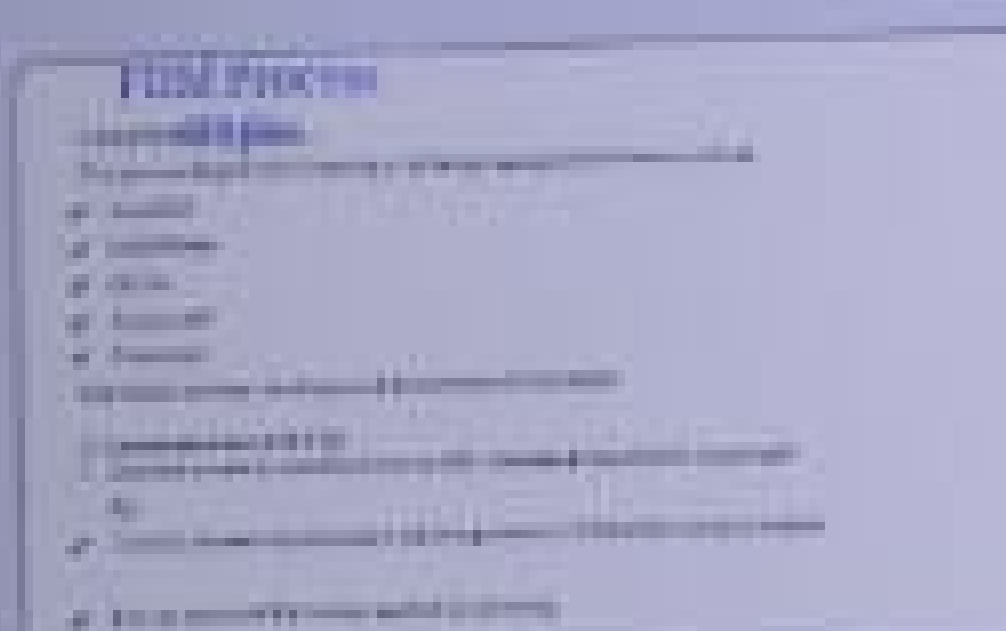
- Divides the model into thin layers
- Generates G-code instructions
- Sets parameters like:
 - ✓ Layer height
 - ✓ Print speed
 - ✓ Temperature
 - ✓ Infill percentage



26



27



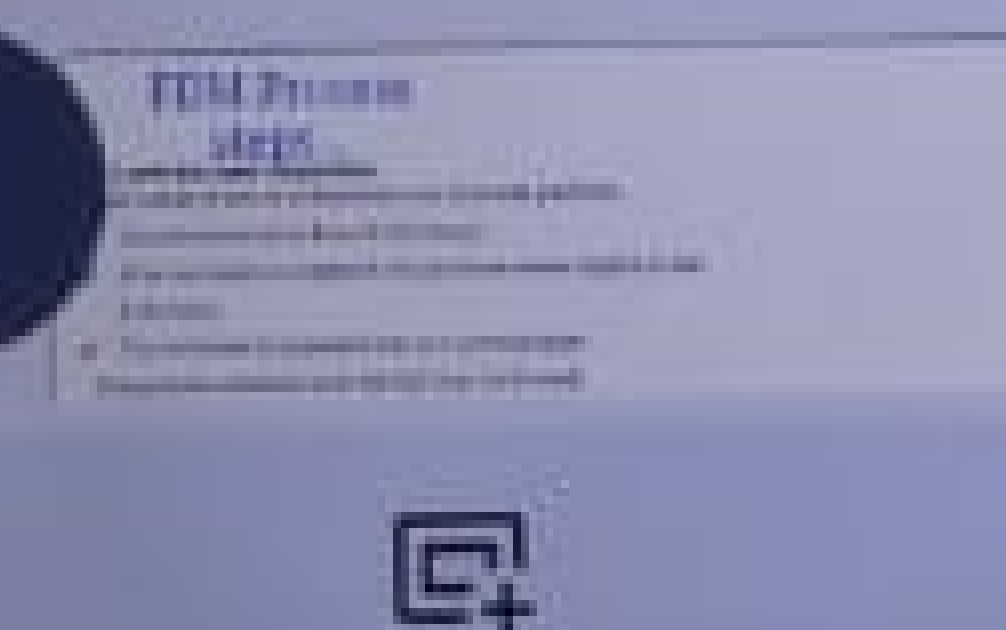
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29



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FDM Process

4. Loading the Filament

A thermoplastic filament spool is loaded into the printer.

Common materials:

- ✓ PLA
- ✓ ABS (Acrylonitrile Butadiene Styrene)
- ✓ PETG (Polyethylene Terephthalate Glycol)
- ✓ TPU (Thermoplastic Polyurethane)
- ✓ Nylon

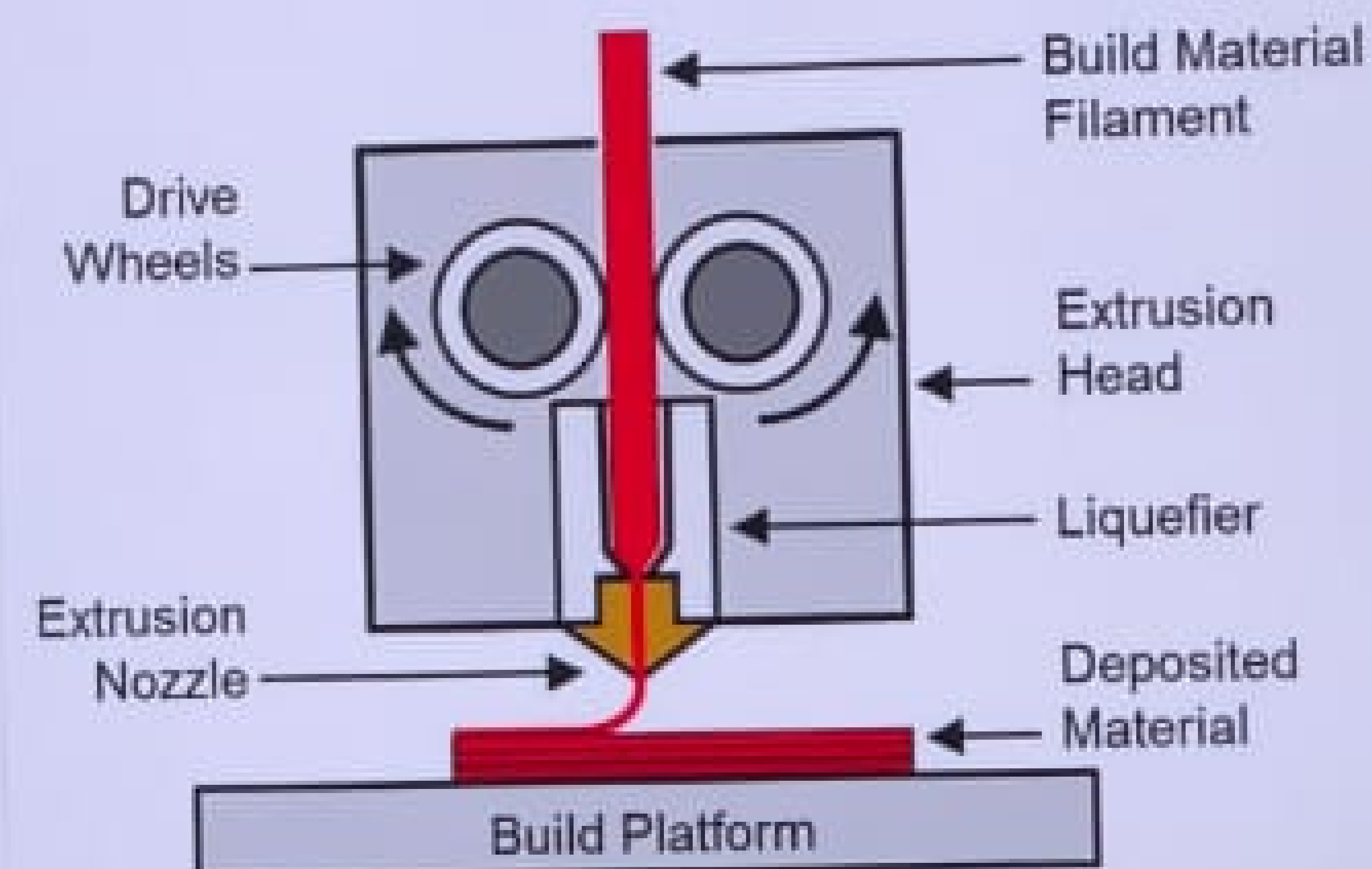
The filament is fed into the extruder.

5. Heating and Extrusion

The nozzle is heated to the required temperature

- ✓ PLA: about 180–220°C
- ✓ ABS: about 220–250°C

The extruder motor pushes the filament into the heated nozzle where it melts.





FDM Process steps...

6. Layer-by-Layer Deposition

The molten material is deposited onto the build platform.

- ✓ The nozzle moves in X and Y directions.
- ✓ After one layer is completed, the platform moves slightly in the Z direction.
- ✓ The next layer is deposited over the previous layer.

This process continues until the full object is formed.

7. Cooling and Solidification

The extruded material cools quickly and solidifies.

Cooling fans help

- ✓ Maintain shape accuracy
- ✓ Improve surface quality
- ✓ Prevent deformation

FDM Process

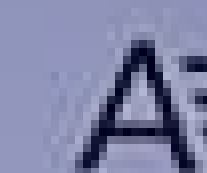
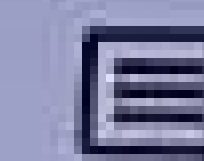
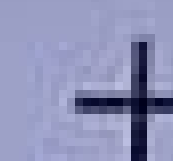
8. Support Structure Removal

For overhanging parts, support materials may be printed.

After printing:

- ✓ Supports are removed manually or chemically.
- ✓ Surface finishing may be done if required.





Materials Used in FDM Technology

- ✓ PLA (**P**oly**L**actic **A**cid)
- ✓ ABS (**A**crylonitrile **B**utadiene **S**tyrene)
- ✓ PETG (**P**oly**E**thylene **T**erephthalate **G**lycol)
- ✓ TPU (**T**hermo**P**lastic **P**olyurethane)
- ✓ ASA (**A**crylonitrile **S**tyrene **A**crylate)



Materials Used in FDM Technology

1. PLA (Polylactic Acid)

- ✓ Easy to print, biodegradable, and produces good surface finish.
- ✓ Commonly used for prototypes, models, and low-load components.
- ✓ **Melting point:** ~150–180°C

2. ABS (Acrylonitrile Butadiene Styrene)

- ✓ Strong, tough, and has good heat resistance.
- ✓ Used for functional parts, housings, and automotive components.
- ✓ **Melting point:** ~200–240°C

3. PETG (Polyethylene Terephthalate Glycol)

- ✓ Strong, durable, and resistant to chemicals and moisture.
- ✓ Suitable for functional parts and mechanical components.
- ✓ **Melting point:** ~230–260°C

4. TPU (Thermoplastic Polyurethane)

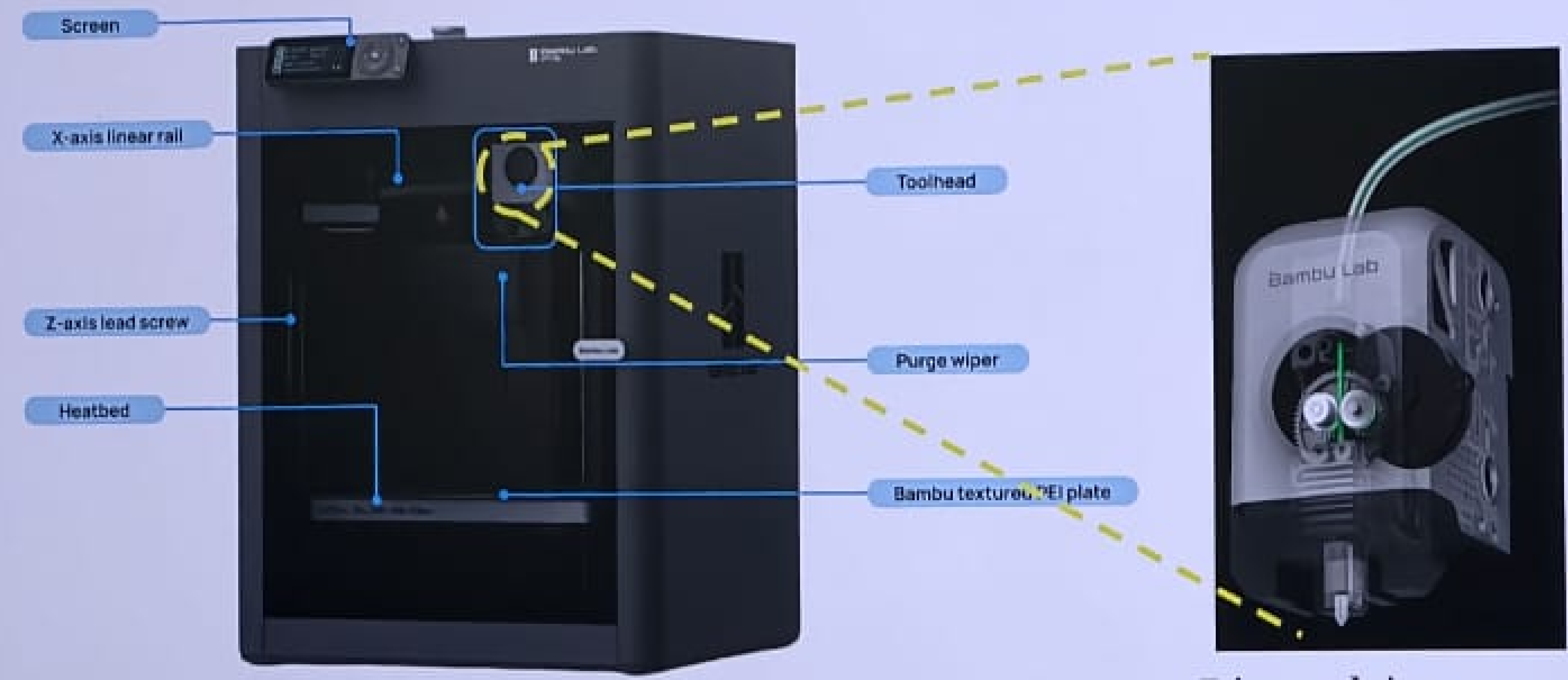
- ✓ Flexible, elastic, and resistant to abrasion.
- ✓ Used for seals, gaskets, flexible parts, and protective covers.
- ✓ **Melting point:** ~170–230°C (depends on TPU grade)

5. ASA (Acrylonitrile Styrene Acrylate)

- ✓ Strong and highly resistant to UV, weather, and heat.
- ✓ Suitable for outdoor and automotive applications.
- ✓ **Melting point:** ~240–260°C



Bambu Lab P1S 3D Printer



Direct-drive Extruder

Wire spools

